Algebra 1 — Basics for Beginners

12-page study guide based on the overall concepts of the YouTube lesson you provided.

Video: Algebra 1 Basics for Beginners — UltimateAlgebra

This is an original study guide, not a transcript of the video.

1. What Is Algebra?

Algebra uses numbers, symbols, and variables to represent quantities and relationships.

Variable: an unknown or changing number, such as x.

Constant: a fixed number, such as 7 in 3x + 7.

Coefficient: the number multiplying a variable. In 5x, it is 5.

Term: a number, variable, or product separated by + or –.

Expression: 3x + 5. **Equation:** 3x + 5 = 17.

Example: If x = 4, then 3x + 5 = 17.

2. Arithmetic & Order of Operations

Use the order: Parentheses → Exponents → Multiplication/Division → Addition/Subtraction.

Example: $3 + 2 \times 5 = 13$.

Distributive property: $a(b + c) = ab + ac$.

Example: $4(x + 3) = 4x + 12$.

Important properties include commutative, associative, and identity properties.

3. Simplifying Expressions

To simplify, remove parentheses when appropriate and combine like terms.

Like terms have the same variable part and exponents.

Example: $3x + 5 – 2x + 7 = x + 12$.

Example: $2(3x – 4) + x = 7x – 8$.

Do not combine unlike terms such as $3x$ and $3x^2$.

4. One-Step Equations

The goal is to isolate the variable using inverse operations.

$x + 7 = 12 \rightarrow x = 5$.

$x - 4 = 9 \rightarrow x = 13$.

$3x = 18 \rightarrow x = 6$.

$x/5 = 4 \rightarrow x = 20$.

Always check the answer in the original equation.

5. Multi-Step Equations

Simplify first, then isolate the variable step by step.

$3x + 5 = 20 \rightarrow 3x = 15 \rightarrow x = 5$.

$2x - 7 = 13 \rightarrow 2x = 20 \rightarrow x = 10$.

$2(x + 3) = 14 \rightarrow 2x + 6 = 14 \rightarrow x = 4$.

Whatever operation is done to one side must also be done to the other.

6. Variables on Both Sides

Move variable terms to one side and constants to the other.

$5x + 2 = 3x + 10 \rightarrow 2x + 2 = 10 \rightarrow x = 4$.

Some equations have no solution; others have infinitely many solutions.

Example: $2x + 4 = 2x + 4$ is true for every x.

7. Fractions, Decimals & Proportions

Fractions can be solved using common denominators or by multiplying by a common denominator.

$$x/3 + 2 = 6 \rightarrow x/3 = 4 \rightarrow x = 12.$$

For proportions $a/b = c/d$, cross multiplication gives $ad = bc$, when denominators are nonzero.

Decimals follow the same algebraic principles; keep place values organized.

8. Exponents & Roots

An exponent shows repeated multiplication: $a^2 = a \times a$.

Rules: a■a■ = a■■■; a■/a■ = a■■■; (a■)■ = a■■.

$\sqrt{49} = 7$.

$x^2 = 25$ gives $x = 5$ or $x = -5$.

Be careful with negative signs and exponent rules.

9. Linear Equations & Graphs

A linear equation represents a straight-line relationship.

Slope-intercept form: $y = mx + b$.

m is the slope and b is the y-intercept.

Slope: m = (y■ – y■)/(x■ – x■).

Example: $y = 2x + 1$ has slope 2 and y-intercept 1.

10. Word Problems

Translate words into mathematical expressions or equations.

“A number increased by 7” → $x + 7$.

“Five times a number” → $5x$.

Method: identify the unknown → define a variable → write an equation → solve → check.

Example: $x + 9 = 20$ → $x = 11$.

11. Common Mistakes

Do not combine unlike terms: $3x + 2$ is not $5x$.

Distribute to every term: $3(x + 4) = 3x + 12$.

Keep both sides of an equation balanced.

Watch negative signs.

Remember $x^2 = 25$ has two real solutions: ± 5.

Check your final answer.

12. Final Review & Practice

Core idea: Algebra is about representing relationships and solving for unknowns.

Practice 1: Simplify $4x + 7 – 2x + 3$. Answer: $2x + 10$.

Practice 2: Solve $x + 8 = 21$. Answer: 13.

Practice 3: Solve $3x – 4 = 17$. Answer: 7.

Practice 4: Solve $2(x + 5) = 24$. Answer: 7.

Practice 5: Find the slope through (1,3) and (3,7). Answer: 2.

Practice 6: For $y = –2x + 5$, give slope and intercept. Answer: –2 and 5.

Practice 7: Solve $x^2 = 36$. Answer: ±6.